

Chiseled Bookshelf redstone behavior is inconsistent

MC-257178 Redstone

Status	Resolved / Fixed
Affected	22w44a, 1.19.3
Fixed in	23w16a
Reported	2022-11-03

STEPS TO REPRODUCE

1. Launch Minecraft 22w44a.
2. Create a Creative world with default settings.
3. Place a chiseled bookshelf over a hopper, and set up a comparator to visually measure the bookshelf's signal strength.
4. Power the hopper.
5. Fill all hopper slots with plain books.
6. Insert books into the bookshelf in this order: Protection I enchanted book, plain book, plain book, Protection II enchanted book, Protection III enchanted book, and Protection IV enchanted book.
7. De-power the hopper.
8. Observe the comparator output as the two plain books are removed: it indicates 6 after the first removal, then 5 and 4 after the second removal; the bookshelf texture does not differ from when slots 5 and 6 are empty.
9. Add two books-and-quills to the bookshelf, one at a time, and observe the comparator output indicate 5 and then 6.
10. Remove all books from the bookshelf and observe that they are retrieved in this order: Protection IV enchanted book, Protection III enchanted book, Protection II enchanted book, book-and-quill, book-and-quill, and Protection I enchanted book.

OBSERVED BEHAVIOUR

With the hopper powered, it removes the books from the chiseled bookshelf's occupied slots 2 and 3 rather than shifting the remaining books toward the first available slots. After the first book is removed, the comparator output temporarily reads 6; while the second book is removed, it reads 5 and then settles at 4. The bookshelf's texture does not visually distinguish this state from one in which slots 5 and 6 are empty. When the hopper is powered again and two books-and-quills are added, they are inserted into the newly empty slots 2 and 3, and the comparator output increases from 4 to 5 and then 6. Removing all six books afterward returns them in the order Protection IV, Protection III, Protection II, book-and-quill, book-and-quill, and Protection I, confirming that the books-and-quills occupied slots 2 and 3 while the original books remained in their existing slots. This produces a mismatch between the bookshelf's slot-level insertion and removal behavior, its comparator output, and its unchanged visual appearance.

EXPECTED BEHAVIOUR

The comparator should consistently report the slot most recently interacted with, rather than producing signals corresponding to unrelated or shifted slots. After the hopper removes the books from slots 2 and 3, the output should reflect those slots in sequence, and inserting books-and-quills into slots 2 and 3 should likewise produce signals for slots 2 and 3. The bookshelf's texture should also clearly show which slots are empty.

ENVIRONMENT

Minecraft Java Edition Versions: 22w44a, 1.19.3 (includes snapshots)

ORIGINAL DESCRIPTION

Now that hopper can interact with chiseled bookshelves in 22w44a, the redstone behavior no longer matches the intended behavior from the announcement.

Setup

- Place a chiseled bookshelf over a hopper. Set up a comparator to visually measure the signal strength from the bookshelf.
- Power the hopper.
- Fill all hopper slots with plain books, so it can take in only more plain books.
- Insert books into the bookshelf in the following order: Protection I enchanted book, plain book, plain book, Protection II enchanted book, Protection III enchanted book, Protection IV enchanted book.
- De-power the hopper. Observe comparator output as the two plain books are removed.
- Add two books-and-quills to the chiseled bookshelf, observing comparator output as this is done.
- Remove the books from the bookshelf.

Actual behavior

At step 5, the books are removed from slots 2 and 3. Comparator output indicates 6 after the first removal, then 5 and then 4 after the second. Block texture is no different from when slots 5 and 6 were empty.

At step 6, the books-and-quills are inserted into slots 2 and 3. Comparator output indicates 5 then 6.

At step 7, we get back the books in the following order: Prot IV, Prot III, Prot II, book-and-quill, book-and-quill, Prot I, confirming that the two books-and-quills were inserted into slots 2 and 3.

Expected behavior

There are several ways this could work to be consistent:

- The comparator output indicates slots 2 and 3 when slots 2 and 3 are the last interacted with, as it is currently defined. And ideally the block texture would visually indicate which slots are empty too.
- The hopper works more like "remove book from slot 2 then shift all books over" so the empty slots are always at the end. In that case the comparator output should have been 5 (not 6) after the first removal (since the last action was moving the book in slot 6 to slot 5), the output after the second removal should have immediately been 4 (not 5-then-4), and the order the books are retrieved in step 7 should reflect the books-and-quills having been inserted into slots 5 and 6.
- The hopper can only take from the last filled slot. Although

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-257178>